

KARTHIK RAGULA

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EXPERIENCE

Associate Software Engineer

June 2025 – Present

WaveMaker India Private Ltd, Hyderabad

- Built **20+ Java/Spring MCP tools** that expose WaveMaker platform APIs as LLM-callable operations, enabling AI-driven development workflows across **250+ projects**.
- Engineered **hash-based optimistic concurrency control** across 3 file-editing tools using range-scoped content hashes and per-session agent state, preventing stale parallel edits from silently overwriting changes while avoiding repeated full-file reads.
- Built **6 LLM-driven file-system tools** for code exploration and editing, integrating ripgrep with configurable search filters, glob patterns, context controls, project-root scoping, and a **100-result cap** to bound LLM context and token consumption.
- Converted the database agent into a modular **Database Skill** and implemented **8 MCP tools** for database discovery and HQL query CRUD operations, enabling on-demand database capabilities without unnecessary cross-agent context sharing.
- Built **15+ MCP tools** for internationalization (i18n), enabling AI-assisted localization and multilingual application development.
- Worked within AIRA's **LangGraph-based multi-agent architecture**, integrating capabilities across **7 specialized agents** for UI, backend, vision, screenshot-to-code, mobile, and code-review workflows.
- Contributed to AIRA's **LLM security and context-management architecture**, including bidirectional redaction of API keys and passwords before LLM processing and context compaction for sessions approaching the **200K-token limit**.

SKILLS

Programming	Java, Python, SQL
Backend	Spring Boot, Spring MVC, Spring AI, REST APIs, Microservices
AI Engineering	Agentic AI, LLMs, MCP, LangGraph, RAG, Embeddings, Vector Search
Databases	MySQL, PostgreSQL, HQL
Frontend	ReactJS, HTML5, CSS3
Tools	Git, GitHub, Docker, Maven, JIRA, Ripgrep
DSA	650+ LeetCode 450+ Coding Ninjas (Master Badge) 150+ GeeksforGeeks

EDUCATION

Bachelor of Technology in Computer Science and Engineering

2021 – 2025

Sree Vidyanikethan Engineering College (**9.4/10.0**)

ACHIEVEMENTS

- Solved **1,250+ Data Structures & Algorithms** problems across LeetCode, Coding Ninjas, and GeeksforGeeks.
- Grew a **YouTube** channel to **10,000+ subscribers** and **15M+ views**, creating interview preparation, career guidance, and educational content.
- Served as **DATA Club Secretary** at SVEC, leading a **600+ member** student community and organizing technical workshops, coding events, and career guidance sessions.
- Secured **2nd place in MOOCGATE** and **3rd place in Career Quest** among **1,200+ participants** at ISTE technical competitions.